

Matthew Rizky Hartadi

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SUMMARY

Computer Science graduate with hands-on experience in *full-stack development, AI integration, and scalable backend systems*. Built production-ready applications using **Next.js, React, NestJS, Django REST Framework, PostgreSQL, Prisma, and AWS**. Strong experience in **RAG systems, vector search**, API integration, and performance optimization. I like solving problems, working with people, and building things that feel practical and polished. Away from the laptop, I spent over 10 years in **taekwondo**, competed internationally, taught students of all ages, and now stay active through **triathlon training**.

EDUCATION

Bachelor of Computer Science (Double Degree) – *Graduated 2025*

University of Queensland – Major in **Data Science** – GPA: 6.25/7

University of Indonesia – GPA: 3.59/4

Relevant Coursework: **Machine Learning, Statistical Modelling, Database Systems, Software Engineering, Human-Computer Interaction, Data Structure and Algorithm**

IBM Quantum

Basics of Quantum Information Certificate - *February 2025*

SKILLS

Programming Languages: TypeScript, JavaScript, Python, Java, SQL, Kotlin, Dart

Frontend: Next.js, React, Tailwind CSS, Bootstrap

Backend: NestJS, Django REST Framework, Spring Boot

Database and Cloud: PostgreSQL, Prisma, pgvector, AWS, S3, Supabase

AI and Data: OpenAI, Gemini, RAG, LLM integration, vector search

Tools: Git, VS Code, IntelliJ, Android Studio

Languages: Indonesian, English

PROJECTS

Mandareen - AI-Powered Mandarin Learning SaaS

- **Built** a full-stack Mandarin learning SaaS with **Next.js, NestJS, PostgreSQL, Prisma, and Supabase**, supporting dynamic lesson generation, real-time chat, and structured curriculum delivery.
- **Integrated OpenAI, Gemini embeddings, STT, and TTS** to support **semantic search, speech interaction, and contextual grammar explanations**.
- **Designed and deployed** the system using **Vercel, Railway, Supabase, and S3** for scalable frontend, backend, and audio storage infrastructure.
- **Optimized RAG-based retrieval** and processing performance through **LRU caching, batched operations, parallel lookups, and HNSW vector indexing**, **reducing** query volume by **99.8%** and processing time from **5.3s** to **0.37s**.

MealU - Meal Kit Platform for Students

- **Developed** a full-stack **meal kit ordering platform** for university students, enabling customizable meal selection, recipe browsing, and streamlined delivery.
- Built a mobile-first interface using **React** and **Ionic Framework** to deliver a responsive and accessible user experience.
- Implemented the backend with **Django REST Framework, PostgreSQL, AWS EC2, and S3** for secure API handling, data storage, and image management.
- **Designed** a **QR-based locker delivery system**, allowing contactless and secure pickup while reducing delivery overhead.
- Added community features such as recipe sharing, reviews, and a reward-based credit system to improve engagement and content contribution.

Siasisten Revamp

- **Redesigned** Siasisten (TA management web app) in a **12-person team** under **SCRUM methodology**.
- **Implemented** modern tech stack: **ReactJS (frontend)** and **Django REST framework (backend)**.
- Helped modernize the platform's user experience and improve maintainability through a cleaner full-stack architecture.

ACB-ISBE Web and Mobile App

- Developed responsive **web and mobile applications** using **Flutter** for an international conference with **1,600+ participants**.
- Built a unified experience for event schedules, information access, and cross-platform usability through a shared API and intuitive interface design.

WORK EXPERIENCE

Teaching Assistant of Programming Foundation II - University of Indonesia

Jan 2023 – June 2023

- **Facilitated** weekly Java programming labs for 30+ students, supporting debugging, concept reinforcement, and hands-on practice.
- **Collaborated** with a team of 3 to ensure grading consistency and high-quality feedback.
- **Enhanced** curriculum delivery and student understanding through interactive sessions.

Teaching Assistant of Programming Foundation I - University of Indonesia

August 2022 – Dec 2022

- **Led** weekly **Python programming labs** and created supplemental materials.
- **Streamlined** assessment workflows for 30+ submissions, ensuring fair evaluation.
- Provided one-on-one mentoring to help students build confidence in foundational programming concepts.

AWARDS AND ACHIEVEMENTS

Winner - UQCS Hackathon (2024)

- **Built** UQuizzle, an **AI-powered browser extension** for Echo360 video playback with 4 teammates.
- Used **React** and **Vercel AI SDK** to **generate real-time quizzes** and **detailed lecture notes**, including assignment and exam hints.
- **Won 1st place** out of **30+ teams** and **180+ participants**.

ACTIVITIES

Vice Manager of Partnership - COMPFEST (2022 - 2023)

- Led a **24-member team** and **raised** the equivalent of over **IDR 250 million** in sponsorships through proposal writing and corporate outreach.